

Introduction to Bootstrap Confidence Intervals: Proportion of yellow M&Ms

MATH 213

Group 1

1. Empty out the M&M's package. How many M&M's altogether? How many are yellow? What is the proportion of yellow M&M's in the package?

2/15 yellow = 0.13

2/15 yellow = 0.13

2. If we consider the package to be a RANDOM SAMPLE from the population of ALL M&M's produced by Mars, is the proportion in step 1 a statistic or parameter?

statistic

3. What is the corresponding parameter of interest, in words?

The proportion of yellow m&m's in packs

We are interested in using this “random sample” of M&Ms to find an interval estimate for the population proportion of yellow M&M candies in the population of all M&Ms produced by Mars.

PROBLEM: We don’t have access to the population so we can’t repeatedly sample from it to create a sampling distribution, calculate the 2.5th and 97.5th percentiles and compute the confidence interval.

SOLUTION: We use the technique of BOOTSTRAPPING.

4. There were THREE yellow M&M’s in our package of 16. Follow the steps below

- In your cup of counters, you will need 3 yellows and 13 of another color.
- Put the counters in the cup, shake them up, close your eyes and select a counter. Record it as “yellow” or “not yellow”.
- PUT THE COUNTER you just selected BACK INTO THE CUP.
- Shake them up again and select another. Record it as “yellow” or “not yellow”.
- Keep going until you have selected 16 counters from your cup altogether.
- Record your results in the tables to the right.

Counter	Yellow or NOT
1	Not
2	Not
3	Not
4	Yellow
5	Not
6	Not
7	Not
8	Yellow
9	Not
10	Not
11	Yellow
12	Not
13	Not
14	Yellow
15	Not
16	Not

Counter	Yellow or NOT
1	not
2	not
3	not
4	not
5	not
6	not
7	yellow
8	not
9	not
10	not
11	yellow
12	not
13	not
14	not
15	not
16	not

The sample you generated by repeatedly selecting counters from the cup is called a **bootstrap sample**.

5. Now, calculate the bootstrap statistic (in this case, the bootstrap proportion) by calculating the proportion of yellow in your bootstrap sample. Show your work for all teammates here:

Alexis: $4/16 = 1/4 = .25$

Itunu: $2/16 = 0.125$

6. When you are done, plot your bootstrap proportions on the dotplot at [this link](#).

Now we will go to StatKey

<http://www.lock5stat.com/StatKey/index.html> and do the same thing.

7. In StatKey, go to CI for Single Proportion under Bootstrap Confidence Intervals.

8. Click Edit the Data and input the information for our sample. We need to type 3 for the count and 16 for the sample size.

9. Ask StatKey to generate 1 bootstrap sample. Think about what StatKey has just done behind the scenes. StatKey just created a bootstrap sample just like you did with the counters. What is the bootstrap statistic in this case?

Try this a few more times to get a feel for what is happening.

10. Ask StatKey to generate 2000 more samples. Answer the questions:

11. What shape does the bootstrap distribution have?

Right skew

12. Where is it centered?

.189

13. Check the two tail box in the top left corner of the plot. Change the proportion that appears in the center to 0.95 if necessary so that the 2.5th and 97.5th percentiles are marked.

14. What is our 95% confidence interval for the population proportion of yellow M&Ms produced by the Mars manufacturing process?

0 and .375

15. Complete the sentence of interpretation:

We are 95% confident that of all the M&Ms Mars produces, between 0 and .375 are yellow.

Group 2

1. Empty out the M&M's package. How many M&M's altogether? How many are yellow? What is the proportion of yellow M&M's in the package?

15 total, 1 yellow (proportion of yellow M&M's is 0.067)

2. If we consider the package to be a RANDOM SAMPLE from the population of ALL M&M's produced by Mars, is the proportion in step 1 a statistic or parameter?

A statistic.

3. What is the corresponding parameter of interest, in words?

Proportion of yellow M&M's in the population.

We are interested in using this “random sample” of M&Ms to find an interval estimate for the population proportion of yellow M&M candies in the population of all M&Ms produced by Mars.

PROBLEM: We don’t have access to the population so we can’t repeatedly sample from it to create a sampling distribution, calculate the 2.5th and 97.5th percentiles and compute the confidence interval.

SOLUTION: We use the technique of BOOTSTRAPPING.

4. There were THREE yellow M&M’s in our package of 16. Follow the steps below

- In your cup of counters, you will need 3 yellows and 13 of another color.
- Put the counters in the cup, shake them up, close your eyes and select a counter. Record it as “yellow” or “not yellow”.
- PUT THE COUNTER you just selected BACK INTO THE CUP.
- Shake them up again and select another. Record it as “yellow” or “not yellow”.
- Keep going until you have selected 16 counters from your cup altogether.
- Record your results in the tables to the right.

Counter	Yellow or NOT
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	
14	
15	
16	

Counter	Yellow or NOT
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	
14	
15	
16	

The sample you generated by repeatedly selecting counters from the cup is called a **bootstrap sample**.

5. Now, calculate the bootstrap statistic (in this case, the bootstrap proportion) by calculating the proportion of yellow in your bootstrap sample. Show your work for all teammates here:

6. When you are done, plot your bootstrap proportions on the dotplot at [this link](#).

Now we will go to StatKey

<http://www.lock5stat.com/StatKey/index.html> and do the same thing.

7. In StatKey, go to CI for Single Proportion under Bootstrap Confidence Intervals.

8. Click Edit the Data and input the information for our sample. We need to type 3 for the count and 16 for the sample size.

9. Ask StatKey to generate 1 bootstrap sample. Think about what StatKey has just done behind the scenes. StatKey just created a bootstrap sample just like you did with the counters. What is the bootstrap statistic in this case?

Try this a few more times to get a feel for what is happening.

10. Ask StatKey to generate 2000 more samples. Answer the questions:

11. What shape does the bootstrap distribution have?

12. Where is it centered?

13. Check the two tail box in the top left corner of the plot. Change the proportion that appears in the center to 0.95 if necessary so that the 2.5th and 97.5th percentiles are marked.

14. What is our 95% confidence interval for the population proportion of yellow M&Ms produced by the Mars manufacturing process?

15. Complete the sentence of interpretation:

We are 95% confident that of all the M&Ms Mars produces, between _____ and _____ are yellow.

Group 3

1. Empty out the M&M's package. How many M&M's altogether? How many are yellow? What is the proportion of yellow M&M's in the package?

29 total, 9 yellows. 31% yellow.

2. If we consider the package to be a RANDOM SAMPLE from the population of ALL M&M's produced by Mars, is the proportion in step 1 a statistic or parameter?

Statistic

3. What is the corresponding parameter of interest, in words?

Proportion of yellow m&ms.

We are interested in using this “random sample” of M&Ms to find an interval estimate for the population proportion of yellow M&M candies in the population of all M&Ms produced by Mars.

PROBLEM: We don’t have access to the population so we can’t repeatedly sample from it to create a sampling distribution, calculate the 2.5th and 97.5th percentiles and compute the confidence interval.

SOLUTION: We use the technique of BOOTSTRAPPING.

4. There were THREE yellow M&M’s in our package of 16. Follow the steps below

- In your cup of counters, you will need 3 yellows and 13 of another color.
- Put the counters in the cup, shake them up, close your eyes and select a counter. Record it as “yellow” or “not yellow”.
- PUT THE COUNTER you just selected BACK INTO THE CUP.
- Shake them up again and select another. Record it as “yellow” or “not yellow”.
- Keep going until you have selected 16 counters from your cup altogether.
- Record your results in the tables to the right.

Counter	Yellow or NOT
1	N n
2	Y n
3	N y
4	N n
5	N n
6	N n
7	N n
8	nn
9	nn
10	nn
11	nn
12	yy
13	yn
14	ny
15	nn
16	nn

Counter	Yellow or NOT
1	n
2	n
3	y
4	n
5	n
6	n
7	n
8	n
9	n
10	n
11	n
12	n
13	n
14	n
15	n
16	n

The sample you generated by repeatedly selecting counters from the cup is called a **bootstrap sample**.

5. Now, calculate the bootstrap statistic (in this case, the bootstrap proportion) by calculating the proportion of yellow in your bootstrap sample. Show your work for all teammates here:

$$3/16 = 0.1875$$

$$3/16 = 0.1875$$

$$1/16 = 0.0625$$

6. When you are done, plot your bootstrap proportions on the dotplot at [this link](#).

Now we will go to StatKey

<http://www.lock5stat.com/StatKey/index.html> and do the same thing.

7. In StatKey, go to CI for Single Proportion under Bootstrap Confidence Intervals.

8. Click Edit the Data and input the information for our sample. We need to type 3 for the count and 16 for the sample size.

9. Ask StatKey to generate 1 bootstrap sample. Think about what StatKey has just done behind the scenes. StatKey just created a bootstrap sample just like you did with the counters. What is the bootstrap statistic in this case?

Try this a few more times to get a feel for what is happening.

10. Ask StatKey to generate 2000 more samples. Answer the questions:

11. What shape does the bootstrap distribution have?

Right skewed

12. Where is it centered?

0.188

13. Check the two tail box in the top left corner of the plot. Change the proportion that appears in the center to 0.95 if necessary so that the 2.5th and 97.5th percentiles are marked.

14. What is our 95% confidence interval for the population proportion of yellow M&Ms produced by the Mars manufacturing process?

(0, .375)

15. Complete the sentence of interpretation:

We are 95% confident that of all the M&Ms Mars produces, between 0 and .375 are yellow.

Group 4

1. Empty out the M&M's package. How many M&M's altogether? How many are yellow? What is the proportion of yellow M&M's in the package?

1 yellow out of 16 total

$$1/16 = .0625$$

2. If we consider the package to be a RANDOM SAMPLE from the population of ALL M&M's produced by Mars, is the proportion in step 1 a statistic or parameter?

Statistic

3. What is the corresponding parameter of interest, in words?

Proportion of all M&Ms that are made that are yellow

We are interested in using this “random sample” of M&Ms to find an interval estimate for the population proportion of yellow M&M candies in the population of all M&Ms produced by Mars.

PROBLEM: We don’t have access to the population so we can’t repeatedly sample from it to create a sampling distribution, calculate the 2.5th and 97.5th percentiles and compute the confidence interval.

SOLUTION: We use the technique of BOOTSTRAPPING.

4. There were THREE yellow M&M’s in our package of 16. Follow the steps below

- In your cup of counters, you will need 3 yellows and 13 of another color.
- Put the counters in the cup, shake them up, close your eyes and select a counter. Record it as “yellow” or “not yellow”.
- PUT THE COUNTER you just selected BACK INTO THE CUP.
- Shake them up again and select another. Record it as “yellow” or “not yellow”.
- Keep going until you have selected 16 counters from your cup altogether.
- Record your results in the tables to the right.

Counter	Yellow or NOT
1	not
2	not
3	not
4	not
5	yellow
6	not
7	yellow
8	not
9	not
10	yellow
11	yellow
12	yellow
13	not
14	yellow
15	not
16	not

Counter	Yellow or NOT
1	not
2	yellow
3	not
4	yellow
5	not
6	yellow
7	not
8	not
9	yellow
10	not
11	not
12	not
13	not
14	not
15	not
16	not

The sample you generated by repeatedly selecting counters from the cup is called a **bootstrap sample**.

5. Now, calculate the bootstrap statistic (in this case, the bootstrap proportion) by calculating the proportion of yellow in your bootstrap sample. Show your work for all teammates here:

$$5/16 = .3125$$

$$4/16 = .25$$

6. When you are done, plot your bootstrap proportions on the dotplot at [this link](#).

Now we will go to StatKey

<http://www.lock5stat.com/StatKey/index.html> and do the same thing.

7. In StatKey, go to CI for Single Proportion under Bootstrap Confidence Intervals.

8. Click Edit the Data and input the information for our sample. We need to type 3 for the count and 16 for the sample size.

9. Ask StatKey to generate 1 bootstrap sample. Think about what StatKey has just done behind the scenes. StatKey just created a bootstrap sample just like you did with the counters. What is the bootstrap statistic in this case?

0.125

Try this a few more times to get a feel for what is happening.

10. Ask StatKey to generate 2000 more samples. Answer the questions:

11. What shape does the bootstrap distribution have?

Right skew

12. Where is it centered?

.188

13. Check the two tail box in the top left corner of the plot. Change the proportion that appears in the center to 0.95 if necessary so that the 2.5th and 97.5th percentiles are marked.

14. What is our 95% confidence interval for the population proportion of yellow M&Ms produced by the Mars manufacturing process?

(0,.375)

15. Complete the sentence of interpretation:

We are 95% confident that of all the M&Ms Mars produces, between 0 and .375 are yellow.

Group 5

1. Empty out the M&M's package. How many M&M's altogether? How many are yellow? What is the proportion of yellow M&M's in the package?

15 total, 2 yellow

$2/15 = 0.133$

2. If we consider the package to be a RANDOM SAMPLE from the population of ALL M&M's produced by Mars, is the proportion in step 1 a statistic or parameter?

Statistic

3. What is the corresponding parameter of interest, in words?

Proportion of yellow M&M's in a package

We are interested in using this “random sample” of M&Ms to find an interval estimate for the population proportion of yellow M&M candies in the population of all M&Ms produced by Mars.

PROBLEM: We don’t have access to the population so we can’t repeatedly sample from it to create a sampling distribution, calculate the 2.5th and 97.5th percentiles and compute the confidence interval.

SOLUTION: We use the technique of BOOTSTRAPPING.

4. There were THREE yellow M&M’s in our package of 16. Follow the steps below

- In your cup of counters, you will need 3 yellows and 13 of another color.
- Put the counters in the cup, shake them up, close your eyes and select a counter. Record it as “yellow” or “not yellow”.
- PUT THE COUNTER you just selected BACK INTO THE CUP.
- Shake them up again and select another. Record it as “yellow” or “not yellow”.
- Keep going until you have selected 16 counters from your cup altogether.
- Record your results in the tables to the right.

Counter	Yellow or NOT
1	NOT
2	YELLOW
3	NOT
4	NOT
5	NOT
6	YELLOW
7	NOT
8	NOT
9	NOT
10	NOT
11	YELLOW
12	YELLOW
13	YELLOW
14	NOT
15	NOT
16	YELLOW

Counter	Yellow or NOT
1	YELLOW
2	YELLOW
3	NOT
4	NOT
5	YELLOW
6	NOT
7	NOT
8	NOT
9	NOT
10	NOT
11	NOT
12	NOT
13	NOT
14	NOT
15	NOT
16	NOT

Counter	Yellow or NOT
1	NOT
2	NOT
3	NOT
4	YELLOW
5	NOT
6	NOT
7	NOT
8	NOT
9	NOT
10	NOT
11	NOT
12	NOT
13	NOT
14	YELLOW
15	NOT
16	NOT

The sample you generated by repeatedly selecting counters from the cup is called a **bootstrap sample**.

5. Now, calculate the bootstrap statistic (in this case, the bootstrap proportion) by calculating the proportion of yellow in your bootstrap sample. Show your work for all teammates here:

$$3/16 = 0.1875$$

$$2/16 = 0.125$$

$$6/16 = 0.375$$

6. When you are done, plot your bootstrap proportions on the dotplot at [this link](#).

Now we will go to StatKey

<http://www.lock5stat.com/StatKey/index.html> and do the same thing.

7. In StatKey, go to CI for Single Proportion under Bootstrap Confidence Intervals.

8. Click Edit the Data and input the information for our sample. We need to type 3 for the count and 16 for the sample size.

9. Ask StatKey to generate 1 bootstrap sample. Think about what StatKey has just done behind the scenes. StatKey just created a bootstrap sample just like you did with the counters. What is the bootstrap statistic in this case?

0.188 a proportion of yellows in a sample

Try this a few more times to get a feel for what is happening.

10. Ask StatKey to generate 2000 more samples. Answer the questions:

11. What shape does the bootstrap distribution have?

Right skewed

12. Where is it centered?

0.185

13. Check the two tail box in the top left corner of the plot. Change the proportion that appears in the center to 0.95 if necessary so that the 2.5th and 97.5th percentiles are marked.

14. What is our 95% confidence interval for the population proportion of yellow M&Ms produced by the Mars manufacturing process?

(0, 0.375)

15. Complete the sentence of interpretation:

We are 95% confident that of all the M&Ms Mars produces, between 0 and 0.375 are yellow.

Group 6

1. Empty out the M&M's package. How many M&M's altogether? How many are yellow? What is the proportion of yellow M&M's in the package?

There are 13 total M&M's. 2 are yellow. 15.38% of the M&M's are yellow.

2. If we consider the package to be a RANDOM SAMPLE from the population of ALL M&M's produced by Mars, is the proportion in step 1 a statistic or parameter?

The proportion in step 1 would be a statistic.

3. What is the corresponding parameter of interest, in words?

All of the M&M's produced by Mars.

We are interested in using this “random sample” of M&Ms to find an interval estimate for the population proportion of yellow M&M candies in the population of all M&Ms produced by Mars.

PROBLEM: We don’t have access to the population so we can’t repeatedly sample from it to create a sampling distribution, calculate the 2.5th and 97.5th percentiles and compute the confidence interval.

SOLUTION: We use the technique of BOOTSTRAPPING.

4. There were THREE yellow M&M’s in our package of 16. Follow the steps below

- In your cup of counters, you will need 3 yellows and 13 of another color.
- Put the counters in the cup, shake them up, close your eyes and select a counter. Record it as “yellow” or “not yellow”.
- PUT THE COUNTER you just selected BACK INTO THE CUP.
- Shake them up again and select another. Record it as “yellow” or “not yellow”.
- Keep going until you have selected 16 counters from your cup altogether.
- Record your results in the tables to the right.

Counter	Yellow or NOT
1	Not
2	Yellow
3	Not
4	Not
5	Not
6	Not
7	Yellow
8	Not
9	Not
10	Not
11	Not
12	Not
13	Yellow
14	Not
15	Not
16	Not

Counter	Yellow or NOT
1	Not
2	Not
3	Yellow
4	Not
5	Not
6	Not
7	Not
8	Yellow
9	Not
10	Not
11	Not
12	Not
13	Not
14	Not
15	Not
16	Not

The sample you generated by repeatedly selecting counters from the cup is called a **bootstrap sample**.

5. Now, calculate the bootstrap statistic (in this case, the bootstrap proportion) by calculating the proportion of yellow in your bootstrap sample. Show your work for all teammates here:

2 out of 16 were yellow. Therefore, 0.125 is the proportion of yellow in our bootstrap.

6. When you are done, plot your bootstrap proportions on the dotplot at [this link](#).

Now we will go to StatKey

<http://www.lock5stat.com/StatKey/index.html> and do the same thing.

7. In StatKey, go to CI for Single Proportion under Bootstrap Confidence Intervals.

8. Click Edit the Data and input the information for our sample. We need to type 3 for the count and 16 for the sample size.

9. Ask StatKey to generate 1 bootstrap sample. Think about what StatKey has just done behind the scenes. StatKey just created a bootstrap sample just like you did with the counters. What is the bootstrap statistic in this case?

0.188 is the bootstrap statistic.

Try this a few more times to get a feel for what is happening.

10. Ask StatKey to generate 2000 more samples. Answer the questions:

11. What shape does the bootstrap distribution have?

There is a right skew in the bootstrap distribution and it has a mean of 0.189

12. Where is it centered?

0.189

13. Check the two tail box in the top left corner of the plot. Change the proportion that appears in the center to 0.95 if necessary so that the 2.5th and 97.5th percentiles are marked.

14. What is our 95% confidence interval for the population proportion of yellow M&Ms produced by the Mars manufacturing process?

0.000 and 0.375

15. Complete the sentence of interpretation:

We are 95% confident that of all the M&Ms Mars produces, between 0 and .375 are yellow.

Group 7

1. Empty out the M&M's package. How many M&M's altogether? How many are yellow? What is the proportion of yellow M&M's in the package?

15 total, 0 yellow = 0.00

16 total, 2 yellow = 0.0125

31 total, 2 yellow = 0.065

2. If we consider the package to be a RANDOM SAMPLE from the population of ALL M&M's produced by Mars, is the proportion in step 1 a statistic or parameter?

statistic

3. What is the corresponding parameter of interest, in words?

p-hat

We are interested in using this “random sample” of M&Ms to find an interval estimate for the population proportion of yellow M&M candies in the population of all M&Ms produced by Mars.

PROBLEM: We don’t have access to the population so we can’t repeatedly sample from it to create a sampling distribution, calculate the 2.5th and 97.5th percentiles and compute the confidence interval.

SOLUTION: We use the technique of BOOTSTRAPPING.

4. There were THREE yellow M&M’s in our package of 16. Follow the steps below

- In your cup of counters, you will need 3 yellows and 13 of another color.
- Put the counters in the cup, shake them up, close your eyes and select a counter. Record it as “yellow” or “not yellow”.
- PUT THE COUNTER you just selected BACK INTO THE CUP.
- Shake them up again and select another. Record it as “yellow” or “not yellow”.
- Keep going until you have selected 16 counters from your cup altogether.
- Record your results in the tables to the right.

Counter	Yellow or NOT
1	N
2	N
3	Y
4	Y
5	N
6	N
7	N
8	N
9	Y
10	N
11	N
12	Y
13	Y
14	Y
15	N
16	Y 7/16 = 0.4375

Counter	Yellow or NOT
1	N
2	N
3	N
4	N
5	N
6	N
7	N
8	N
9	N
10	N
11	Y
12	N
13	N
14	N
15	N
16	N 1/16 = 0.0625

The sample you generated by repeatedly selecting counters from the cup is called a **bootstrap sample**.

5. Now, calculate the bootstrap statistic (in this case, the bootstrap proportion) by calculating the proportion of yellow in your bootstrap sample. Show your work for all teammates here:

B1:0.4375

B2:0.0625

6. When you are done, plot your bootstrap proportions on the dotplot at [this link](#).

Now we will go to StatKey

<http://www.lock5stat.com/StatKey/index.html> and do the same thing.

7. In StatKey, go to CI for Single Proportion under Bootstrap Confidence Intervals.

8. Click Edit the Data and input the information for our sample. We need to type 3 for the count and 16 for the sample size.

9. Ask StatKey to generate 1 bootstrap sample. Think about what StatKey has just done behind the scenes. StatKey just created a bootstrap sample just like you did with the counters. What is the bootstrap statistic in this case?

0.25

Try this a few more times to get a feel for what is happening.

10. Ask StatKey to generate 2000 more samples. Answer the questions:

11. What shape does the bootstrap distribution have?

Right skewed graph

12. Where is it centered?

0.188

13. Check the two tail box in the top left corner of the plot. Change the proportion that appears in the center to 0.95 if necessary so that the 2.5th and 97.5th percentiles are marked.

14. What is our 95% confidence interval for the population proportion of yellow M&Ms produced by the Mars manufacturing process?

(0.000, 0.375)

15. Complete the sentence of interpretation:

We are 95% confident that of all the M&Ms Mars produces, between 0% and 37.5% are yellow.

Group 8

1. Empty out the M&M's package. How many M&M's altogether? How many are yellow? What is the proportion of yellow M&M's in the package?

13, There are 3 yellow, The proportions is .23

2. If we consider the package to be a RANDOM SAMPLE from the population of ALL M&M's produced by Mars, is the proportion in step 1 a statistic or parameter?

Statistic

3. What is the corresponding parameter of interest, in words?

How many yellow m&m's come in packages

We are interested in using this “random sample” of M&Ms to find an interval estimate for the population proportion of yellow M&M candies in the population of all M&Ms produced by Mars.

PROBLEM: We don’t have access to the population so we can’t repeatedly sample from it to create a sampling distribution, calculate the 2.5th and 97.5th percentiles and compute the confidence interval.

SOLUTION: We use the technique of BOOTSTRAPPING.

4. There were THREE yellow M&M’s in our package of 16. Follow the steps below

- In your cup of counters, you will need 3 yellows and 13 of another color.
- Put the counters in the cup, shake them up, close your eyes and select a counter. Record it as “yellow” or “not yellow”.
- PUT THE COUNTER you just selected BACK INTO THE CUP.
- Shake them up again and select another. Record it as “yellow” or “not yellow”.
- Keep going until you have selected 16 counters from your cup altogether.
- Record your results in the tables to the right.

Counter	Yellow or NOT
1	Not
2	Yellow
3	Yellow
4	Not
5	Not
6	Not
7	Not
8	Not
9	Not
10	Not
11	Not
12	Not
13	Not
14	Not
15	Not
16	Yellow

Counter	Yellow or NOT
1	Not
2	Yellow
3	Not
4	Not
5	not
6	not
7	not
8	not
9	not
10	not
11	yellow
12	not
13	not
14	not
15	not
16	not

Counter	Yellow or NOT
1	no
2	no
3	no
4	n0
5	no
6	yellow
7	yellow
8	yellow
9	yellow
10	no
11	n0
12	no
13	no
14	no
15	no
16	no

The sample you generated by repeatedly selecting counters from the cup is called a **bootstrap sample**.

5. Now, calculate the bootstrap statistic (in this case, the bootstrap proportion) by calculating the proportion of yellow in your bootstrap sample. Show your work for all teammates here:

Jose: $3/16 = .1875$

Jonah: $2/16 = .125$

Landon: $4/16 = 0.25$

6. When you are done, plot your bootstrap proportions on the dotplot at [this link](#).

Now we will go to StatKey

<http://www.lock5stat.com/StatKey/index.html> and do the same thing.

7. In StatKey, go to CI for Single Proportion under Bootstrap Confidence Intervals.

8. Click Edit the Data and input the information for our sample. We need to type 3 for the count and 16 for the sample size.

9. Ask StatKey to generate 1 bootstrap sample. Think about what StatKey has just done behind the scenes. StatKey just created a bootstrap sample just like you did with the counters. What is the bootstrap statistic in this case?

Try this a few more times to get a feel for what is happening.

10. Ask StatKey to generate 2000 more samples. Answer the questions:

11. What shape does the bootstrap distribution have?

The shape of our graph of M&Ms is a right skew, most of the points are on the left side of 97.5 quantile

12. Where is it centered?

.189

13. Check the two tail box in the top left corner of the plot. Change the proportion that appears in the center to 0.95 if necessary so that the 2.5th and 97.5th percentiles are marked.

14. What is our 95% confidence interval for the population proportion of yellow M&Ms produced by the Mars manufacturing process?

$\hat{P} = .1875$

15. Complete the sentence of interpretation:

We are 95% confident that of all the M&Ms Mars produces, between 0 and 0.375 are yellow.

Group 9

1. Empty out the M&M's package. How many M&M's altogether? How many are yellow? What is the proportion of yellow M&M's in the package?

3/15 M&Ms were yellow

2. If we consider the package to be a RANDOM SAMPLE from the population of ALL M&M's produced by Mars, is the proportion in step 1 a statistic or parameter?

The proportion in step 1 is a statistic, since that one package of M&M's is a sample.

3. What is the corresponding parameter of interest, in words?

The parameter of interest is the proportion of M&M's in a population of packages.

We are interested in using this “random sample” of M&Ms to find an interval estimate for the population proportion of yellow M&M candies in the population of all M&Ms produced by Mars.

PROBLEM: We don’t have access to the population so we can’t repeatedly sample from it to create a sampling distribution, calculate the 2.5th and 97.5th percentiles and compute the confidence interval.

SOLUTION: We use the technique of BOOTSTRAPPING.

4. There were THREE yellow M&M’s in our package of 16. Follow the steps below

- In your cup of counters, you will need 3 yellows and 13 of another color.
- Put the counters in the cup, shake them up, close your eyes and select a counter. Record it as “yellow” or “not yellow”.
- PUT THE COUNTER you just selected BACK INTO THE CUP.
- Shake them up again and select another. Record it as “yellow” or “not yellow”.
- Keep going until you have selected 16 counters from your cup altogether.
- Record your results in the tables to the right.

Counter	Yellow or NOT
1	not
2	not
3	not
4	not
5	not
6	not
7	not
8	not
9	not
10	not
11	not
12	not
13	not
14	not
15	not
16	yellow

Counter	Yellow or NOT
1	not
2	yellow
3	not
4	not
5	not
6	not
7	yellow
8	not
9	not
10	not
11	not
12	yellow
13	not
14	yellow
15	not
16	not

The sample you generated by repeatedly selecting counters from the cup is called a **bootstrap sample**.

5. Now, calculate the bootstrap statistic (in this case, the bootstrap proportion) by calculating the proportion of yellow in your bootstrap sample. Show your work for all teammates here:

$$5/32=0.15625$$

6. When you are done, plot your bootstrap proportions on the dotplot at [this link](#).

Now we will go to StatKey

<http://www.lock5stat.com/StatKey/index.html> and do the same thing.

7. In StatKey, go to CI for Single Proportion under Bootstrap Confidence Intervals.

8. Click Edit the Data and input the information for our sample. We need to type 3 for the count and 16 for the sample size.

9. Ask StatKey to generate 1 bootstrap sample. Think about what StatKey has just done behind the scenes. StatKey just created a bootstrap sample just like you did with the counters. What is the bootstrap statistic in this case?

Try this a few more times to get a feel for what is happening.

10. Ask StatKey to generate 2000 more samples. Answer the questions:

11. What shape does the bootstrap distribution have?

It has a bell shape with a slight left skew

12. Where is it centered?

It is centered around 0.189

13. Check the two tail box in the top left corner of the plot. Change the proportion that appears in the center to 0.95 if necessary so that the 2.5th and 97.5th percentiles are marked.

14. What is our 95% confidence interval for the population proportion of yellow M&Ms produced by the Mars manufacturing process?

0.00-0.375 is the interval

15. Complete the sentence of interpretation:

We are 95% confident that of all the M&Ms Mars produces, between **0** and **0.375** are yellow.

Group 10

1. Empty out the M&M's package. How many M&M's altogether? How many are yellow? What is the proportion of yellow M&M's in the package?
2. If we consider the package to be a RANDOM SAMPLE from the population of ALL M&M's produced by Mars, is the proportion in step 1 a statistic or parameter?
3. What is the corresponding parameter of interest, in words?

We are interested in using this “random sample” of M&Ms to find an interval estimate for the population proportion of yellow M&M candies in the population of all M&Ms produced by Mars.

PROBLEM: We don’t have access to the population so we can’t repeatedly sample from it to create a sampling distribution, calculate the 2.5th and 97.5th percentiles and compute the confidence interval.

SOLUTION: We use the technique of BOOTSTRAPPING.

4. There were THREE yellow M&M’s in our package of 16. Follow the steps below

- In your cup of counters, you will need 3 yellows and 13 of another color.
- Put the counters in the cup, shake them up, close your eyes and select a counter. Record it as “yellow” or “not yellow”.
- PUT THE COUNTER you just selected BACK INTO THE CUP.
- Shake them up again and select another. Record it as “yellow” or “not yellow”.
- Keep going until you have selected 16 counters from your cup altogether.
- Record your results in the tables to the right.

Counter	Yellow or NOT
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	
14	
15	
16	

Counter	Yellow or NOT
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	
14	
15	
16	

The sample you generated by repeatedly selecting counters from the cup is called a **bootstrap sample**.

5. Now, calculate the bootstrap statistic (in this case, the bootstrap proportion) by calculating the proportion of yellow in your bootstrap sample. Show your work for all teammates here:

6. When you are done, plot your bootstrap proportions on the dotplot at [this link](#).

Now we will go to StatKey

<http://www.lock5stat.com/StatKey/index.html> and do the same thing.

7. In StatKey, go to CI for Single Proportion under Bootstrap Confidence Intervals.

8. Click Edit the Data and input the information for our sample. We need to type 3 for the count and 16 for the sample size.

9. Ask StatKey to generate 1 bootstrap sample. Think about what StatKey has just done behind the scenes. StatKey just created a bootstrap sample just like you did with the counters. What is the bootstrap statistic in this case?

Try this a few more times to get a feel for what is happening.

10. Ask StatKey to generate 2000 more samples. Answer the questions:

11. What shape does the bootstrap distribution have?

12. Where is it centered?

13. Check the two tail box in the top left corner of the plot. Change the proportion that appears in the center to 0.95 if necessary so that the 2.5th and 97.5th percentiles are marked.

14. What is our 95% confidence interval for the population proportion of yellow M&Ms produced by the Mars manufacturing process?

15. Complete the sentence of interpretation:

We are 95% confident that of all the M&Ms Mars produces, between _____ and _____ are yellow.